

मौलाना आजाद राष्ट्रीय प्रौद्योगिकी संस्थान - भोपाल
Maulana Azad National Institute of Technology– Bhopal



Department of Computer Science and Engineering
कंप्यूटर विज्ञान और अभियांत्रिकी विभाग

Session 2025-26

“PROJECT-1 REPORT”

On

**“Optimizing Hybrid CNN-LSTM Architectures for High-Accuracy
Pedestrian Activity Recognition”**

**Submitted In partial Fulfillment for the degree of Bachelor of Technology in
Computer Science and Engineering**

SUBMITTED BY:

Akhilesh Patidar (2311201153)
Akshat Agrawal (2311201181)
Prakhar Sharma (2311201156)
Pratham Raghav (2311201162)

GUIDE BY:

Dr. Vijay Bhaskar Semwal,
Assistant Professor,
C.S.E Department,
MANIT Bhopal

मौलाना आजाद राष्ट्रीय प्रौद्योगिकी संस्थान - भोपाल

Maulana Azad National Institute of Technology– Bhopal



Department of Computer Science and Engineering

कंप्यूटर विज्ञान और अभियांत्रिकी विभाग

DECLARATION

I hereby declare that the work, which is presented in this Project Report, entitled “**Optimizing Hybrid CNN-LSTM Architectures for High-Accuracy Pedestrian Activity Recognition**”, in partial fulfilment of the requirements for the award of the degree, submitted in the **Department of Computer Science and Engineering, Maulana Azad National Institute of Technology, Bhopal**. It is an authentic record of my work carried out from __to under the noble guidance of my guide “**Dr. Vijay Bhaskar Semwal**”. The following project and its report, in part or whole, have not been presented or submitted by me for any purpose in any other institute or organization. I hereby declare that the facts mentioned above are true to the best of our knowledge. In case of any unlikely discrepancy that may possibly occur, I will be the one to take responsibility.

Akhilesh Patidar (2311201153)

Akshat Agrawal (2311201181)

Prakhar Sharma (2311201156)

Pratham Raghav (2311201162)

मौलाना आजाद राष्ट्रीय प्रौद्योगिकी संस्थान - भोपाल
Maulana Azad National Institute of Technology– Bhopal



Department of Computer Science and Engineering
कंप्यूटर विज्ञान और अभियांत्रिकी विभाग

CERTIFICATE

This is to certify that “**Akhilesh Patidar, Akshat Agrawal, Prakhar Sharma and Pratham Raghav**”, student of B.Tech 3rd Year (Computer Science & Engineering), have successfully completed their project " **Optimizing Hybrid CNN-LSTM Architectures for High-Accuracy Pedestrian Activity Recognition** " in partial fulfilment of their Bachelor of Technology in Computer Science & Engineering.

Dr. Vijay Bhaskar Semwal

ACKNOWLEDGEMENT

With due respect, we express our deep sense of gratitude to our respected Guide and project coordinator, **Dr. Vijay Bhaskar Semwal**, Assistant Professor, Department of Computer Science & Engineering, MANIT Bhopal, for his valuable guidance and constant encouragement throughout the course of this project. His expertise in embedded systems and machine learning has been a constant source of inspiration. His rigorous evaluation and constructive criticism were very helpful.

We are also grateful to our respected Director, MANIT Bhopal, for permitting us to utilise all the necessary institute facilities.

We gratefully acknowledge the support of the **Head of Department, CSE, MANIT Bhopal**, for providing the necessary laboratory infrastructure and academic support. We also thank all faculty, staff members, and laboratory attendants of our department for their cooperation and help.

We gratefully acknowledge the **UCI Machine Learning Repository** for making the Human Activity Recognition dataset publicly available, which served as the foundation for our model training and evaluation.

Last but certainly not least, we would like to express our deep appreciation towards our family members and batchmates for providing much-needed support and encouragement.

Place: Bhopal

Date: _____

Akhilesh Patidar (2311201153)

Akshat Agrawal (2311201181)

Prakhar Sharma (2311201156)

Pratham Raghav (2311201162)

ABSTRACT

हिंदी सारांश

मानव गतिविधि पहचान (Human Activity Recognition - HAR) सर्वव्यापी कंप्यूटिंग (ubiquitous computing) में एक मौलिक समस्या है, जिसके स्वास्थ्य सेवा निगरानी, बुजुर्गों की देखभाल, खेल विश्लेषण और मानव-कंप्यूटर संपर्क (HCI) में व्यापक अनुप्रयोग हैं। यह प्रोजेक्ट **Arduino Nano BLE 33 Sense Rev2** माइक्रोकंट्रोलर पर तैनात एक रियल-टाइम HAR सिस्टम प्रस्तुत करता है, जो ऑनबोर्ड **इनर्शियल मेजरमेंट यूनिट (IMU)** संकेतों का उपयोग करके दैनिक जीवन की छह गतिविधियाँ – चलना, सीढ़ियाँ चढ़ना, सीढ़ियाँ उतरना, बैठना, खड़ा होना और लेटना – को वर्गीकृत करता है।

यह सिस्टम मॉडल के प्रशिक्षण (training) और सत्यापन (validation) के लिए **UCI Human Activity Recognition** डेटासेट का उपयोग करता है। 3-अक्षीय एक्सेलेरोमीटर (accelerometer) और 3-अक्षीय जायरोस्कोप (gyroscope) से प्राप्त कच्चे सेंसर संकेतों को प्रीप्रोसेस किया जाता है, जिससे तीन मुख्य समूहों में नौ इनपुट चैनल प्राप्त होते हैं: बाँड़ी एक्सेलरेशन, कुल एक्सेलरेशन और बाँड़ी एंगुलर वेलोसिटी।

इस शोध में चार **डेटा ऑगमेंटेशन (Data Augmentation)** रणनीतियों – बेसलाइन, रोटेशन (सभी अक्ष), रैंडम रोटेशन (30° तक), और एडिटिव गॉसियन नॉइज़ – का मूल्यांकन तीन डीप लर्निंग आर्किटेक्चर पर किया गया है: सिंगल-ब्रांच 1D-CNN-BiLSTM, डुअल-ब्रांच जायरोस्कोप/एक्सेलेरोमीटर मॉडल, ट्रिपल-ब्रांच प्रति-अक्ष (per-axis) मॉडल

परिणामों के अनुसार, **रोटेशन-आधारित ऑगमेंटेशन** सभी आर्किटेक्चर में लगातार उच्चतम सटीकता सुधार प्रदान करता है। सभी नौ चैनलों पर संयुक्त ऑगमेंटेशन के साथ प्रशिक्षित अंतिम अनुकूलित मॉडल, **CNN LSTM HAR**, परीक्षण के दौरान **94.64% की सटीकता** और **0.95 का मैक्रो-एवरेज F1-स्कोर** प्राप्त करता है। अंततः, प्रशिक्षित मॉडल को क्वांटाइज़ (quantised) किया गया है और **TensorFlow Lite Micro** का उपयोग करके **Arduino Nano BLE 33 Sense Rev2** पर तैनात किया गया है, जहाँ पहचानी गई गतिविधियों को एक **RGB LED** के माध्यम से रियल-टाइम में प्रदर्शित किया जाता है।

कीवर्ड: मानव गतिविधि पहचान (HAR), Arduino Nano BLE 33 Sense Rev2, कन्वेन्शनल न्यूरल नेटवर्क (CNN), LSTM, डेटा ऑगमेंटेशन, इनर्शियल मेजरमेंट यूनिट (IMU), एज AI (Edge AI), TensorFlow Lite Micro, RGB LED

ABSTRACT

English Abstract

Human Activity Recognition (HAR) is a fundamental problem in ubiquitous computing with wide-ranging applications in healthcare monitoring, elderly care, sports analytics, and human–computer interaction. This project presents a real-time HAR system deployed on the **Arduino Nano BLE 33 Sense Rev2** microcontroller, which classifies six daily-life activities — Walking, Walking Upstairs, Walking Downstairs, Sitting, Standing, and Laying — using onboard inertial measurement unit (IMU) signals.

The system uses the UCI Human Activity Recognition dataset for training and validation. Raw sensor signals from a 3-axis accelerometer and 3-axis gyroscope are preprocessed to yield nine input channels across three signal groups: body acceleration, total acceleration, and body angular velocity. Four data augmentation strategies — baseline, rotation (all axes), random rotation (up to 30°), and additive Gaussian noise — are evaluated across three deep learning architectures: a single-branch 1D-CNN–BiLSTM, a dual-branch gyroscope/accelerometer model, and a triple-branch per-axis model.

Rotation-based augmentation consistently yields the highest accuracy improvement across all architectures. The final optimised model, `CNN_LSTM_HAR`, trained on all nine channels with combined augmentation, achieves a **test accuracy of 94.64%** and a macro-averaged F1-score of 0.95. The trained model is quantised and deployed on the Arduino Nano BLE 33 Sense Rev2 using TensorFlow Lite Micro, with classified activities indicated in real time via an RGB LED.

Keywords: Human Activity Recognition, Arduino Nano BLE 33 Sense Rev2, Convolutional Neural Network, LSTM, Data Augmentation, Inertial Measurement Unit, Edge AI, TensorFlow Lite Micro, RGB LED.

Contents

Acknowledgement	vii
Abstract	xi
List of Figures	xvii
List of Tables	xix
Abbreviations / Notations / Nomenclature	xxi
1 INTRODUCTION	1
1.1 Background and Motivation	1
1.2 Problem Statement	1
1.3 Objectives	2
1.4 Scope and Limitations	2
1.5 Organisation of the Report	3
2 LITERATURE REVIEW	4
2.1 Classical Machine Learning Approaches	4
2.2 Deep Learning for HAR	4
2.2.1 Convolutional Neural Networks	4
2.2.2 Recurrent and Hybrid Architectures	4
2.2.3 Bidirectional LSTM	5
2.3 Data Augmentation in HAR	5
2.4 Edge Deployment of HAR Models	5
2.5 Research Gap	5
3 DATASET AND DATA ANALYSIS	7
3.1 UCI HAR Dataset	7
3.1.1 Signal Preprocessing	7

3.1.2	Dataset Split	7
3.2	Activity Classes	8
3.3	Exploratory Data Analysis	8
3.3.1	Signal Visualisation	8
3.3.2	Class Distribution	9
3.3.3	System Pipeline Overview	9
4	DATA AUGMENTATION	11
4.1	Motivation	11
4.2	Augmentation Techniques	11
4.2.1	Baseline (No Augmentation)	11
4.2.2	Rotation — All Axes	11
4.2.3	Rotation — Random (up to 30°)	11
4.2.4	Additive Gaussian Noise	12
4.3	Augmentation Comparison Results	12
5	MODEL ARCHITECTURES AND EXPERIMENTS	14
5.1	Input Representation	14
5.2	Model 1: Single-Branch 1D-CNN–BiLSTM	14
5.2.1	Architecture	14
5.2.2	Results	15
5.3	Model 2: Dual-Branch CNN–BiLSTM	16
5.3.1	Architecture	16
5.3.2	Results	17
5.4	Model 3: Triple-Branch Per-Axis CNN–BiLSTM	19
5.4.1	Architecture	19
5.4.2	Results	19
5.5	Architecture Comparison Summary	21
6	FINAL MODEL: CNN_LSTM_HAR	22
6.1	Design Rationale	22
6.2	Model Architecture	22
6.3	Training Configuration	24

6.4	Training Progress	25
6.5	Final Test Results	25
7	HARDWARE DEPLOYMENT	27
7.1	Hardware Overview	27
7.1.1	Arduino Nano BLE 33 Sense Rev2	27
7.1.2	RGB LED	27
7.2	Software Libraries	28
7.3	Deployment Pipeline	28
7.3.1	Model Conversion	28
7.3.2	Inference Procedure	29
7.4	System Integration	29
8	CONCLUSION AND FUTURE WORK	31
8.1	Conclusion	31
8.2	Future Work	32
	References	32

List of Figures

3.1	Overall HAR System Pipeline	10
5.1	Model 1: Single-Branch 1D-CNN–BiLSTM Architecture	14
5.2	Model 1 — Test accuracy comparison across augmentation strategies	15
5.3	Model 1 — Validation accuracy curves for all four augmentation strategies	16
5.4	Model 2: Dual-Branch Gyro/Acc Architecture	17
5.5	Model 2 — Test accuracy comparison across augmentation strategies	18
5.6	Model 2 — Validation accuracy curves for all four augmentation strategies	18
5.7	Model 3: Triple-Branch Per-Axis Architecture	19
5.8	Model 3 — Test accuracy comparison across augmentation strategies	20
5.9	Model 3 — Validation accuracy curves for all four augmentation strategies	21
6.1	Final Model CNN_LSTM_HAR Architecture (Total: 185,030 param- eters)	23
7.1	Model Deployment Pipeline: Training to Edge Inference	28
7.2	Hardware Deployment System Flowchart	30

List of Tables

3.1	UCI HAR Dataset Statistics	8
3.2	Activity Classes and RGB LED Colour Mapping	8
4.1	Augmentation Accuracy Comparison Across All Models	12
5.1	Model 1 Classification Report — Rotation (All Axes)	15
5.2	Model 2 Classification Report — Rotation (All Axes)	17
5.3	Model 3 Classification Report — Rotation (All Axes)	20
5.4	Best Accuracy Comparison Across Exploratory Models	21
6.1	Final Model Layer Summary	24
6.2	Final Model Training Hyperparameters	24
6.3	Training Metrics at Selected Epochs	25
6.4	Final Model Performance on UCI HAR Test Set	25
6.5	Final Model Classification Report — All 6 Activity Classes	26
7.1	Arduino Nano BLE 33 Sense Rev2 Key Specifications	27
7.2	Arduino Firmware Libraries	28

ABBREVIATIONS / NOTATIONS / NOMENCLATURE

Abbreviation	Full Form
ACC	Accelerometer
BLE	Bluetooth Low Energy
BN	Batch Normalisation
CNN	Convolutional Neural Network
CSE	Computer Science & Engineering
DL	Deep Learning
FC	Fully Connected (Dense) Layer
GYRO	Gyroscope
HAR	Human Activity Recognition
IMU	Inertial Measurement Unit
IoT	Internet of Things
LPF	Low-Pass Filter
LSTM	Long Short-Term Memory
LR	Learning Rate
MANIT	Maulana Azad National Institute of Technology
ML	Machine Learning
RGB	Red-Green-Blue (LED)
ReLU	Rectified Linear Unit
TFLite	TensorFlow Lite
UCI	University of California, Irvine
body_acc	Body Acceleration (gravity removed)
body_gyro	Body Angular Velocity
total_acc	Total Acceleration (raw, including gravity)

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Human Activity Recognition (HAR) refers to the computational process of identifying physical activities performed by a person based on data captured by body-worn or ambient sensors. With the rapid proliferation of low-cost microelectromechanical systems (MEMS) sensors and edge-capable microcontrollers, HAR has evolved from a laboratory exercise into a practical technology embedded in consumer wearables, smartphones, and IoT devices.

The primary motivation for this project stems from two converging technological trends. First, deep neural networks — particularly hybrid convolutional and recurrent architectures — have demonstrated state-of-the-art performance on raw IMU time-series data, eliminating the need for hand-engineered features. Second, ultra-low-power microcontrollers such as the **Arduino Nano BLE 33 Sense Rev2**, equipped with onboard IMU sensors and capable of running TensorFlow Lite Micro models, now enable end-to-end intelligence at the edge, making real-time, battery-powered HAR practical.

This project addresses the full pipeline: from dataset analysis and augmentation, through systematic model architecture exploration, to embedded deployment with a human-interpretable RGB LED output.

1.2 Problem Statement

The challenge in HAR lies in accurately classifying activities that share similar sensor signal patterns (e.g., Sitting vs. Standing) while remaining robust to variations in sensor placement, body orientation, and individual gait patterns. Classical approaches relying on statistical feature engineering suffer from limited gener-

alisability. This project investigates whether data augmentation combined with deep learning architectures can close this gap and produce a model lightweight enough for microcontroller deployment.

1.3 Objectives

The specific objectives of this project are:

1. To analyse and visualise the UCI HAR dataset across all signal channels and activity classes.
2. To systematically evaluate four data augmentation techniques across three CNN-LSTM model architectures.
3. To design and train a final optimised model achieving greater than 90% test accuracy.
4. To deploy the trained model on the Arduino Nano BLE 33 Sense Rev2 for real-time inference.
5. To implement a visual feedback mechanism using an RGB LED to indicate the classified activity.

1.4 Scope and Limitations

This project uses the UCI HAR benchmark dataset collected from waist-mounted smartphones, which serves as a proxy for the Arduino Nano BLE 33 Sense Rev2's onboard IMU. The deployed system operates offline (no wireless transmission) and provides activity labels via LED colour coding. Extension to continuous streaming over Bluetooth and incorporation of additional activity classes are identified as future work.

1.5 Organisation of the Report

Chapter 2 reviews related literature. Chapter 3 describes the dataset and data analysis. Chapter 4 presents the data augmentation methodology. Chapter 5 details the model architectures and experimental comparison. Chapter 6 describes the final model and results. Chapter 7 covers hardware deployment. Chapter 8 concludes with future directions.

CHAPTER 2

LITERATURE REVIEW

2.1 Classical Machine Learning Approaches

Early HAR systems relied on handcrafted features extracted from sliding windows of sensor data, fed to classifiers such as Support Vector Machines (SVM), k-Nearest Neighbours (k-NN), and Random Forests. Anguita et al. [1] demonstrated that SVM applied to 561 statistical features extracted from the UCI HAR dataset achieves approximately 89% accuracy. While effective, such pipelines require domain expertise for feature engineering and may not generalise across sensor placements.

2.2 Deep Learning for HAR

2.2.1 Convolutional Neural Networks

Ordóñez and Roggen [6] proposed applying 1D CNNs to raw inertial sensor streams, showing that convolutional filters can automatically learn local temporal patterns without manual feature extraction. Subsequent work demonstrated that stacking multiple CNN layers improves discrimination between activities with similar coarse-level dynamics.

2.2.2 Recurrent and Hybrid Architectures

Inoue et al. [5] showed that LSTM networks capture long-range temporal dependencies in activity sequences, outperforming purely convolutional models on activities such as transitions between Sitting and Standing. The combination of CNN for local feature extraction followed by LSTM for sequence modelling (CNN-LSTM) has become a dominant paradigm in HAR literature [4].

2.2.3 Bidirectional LSTM

Bidirectional LSTM (BiLSTM) processes sequences in both forward and backward directions, enriching the hidden state with future context. Zhang and Sawchuk [12] reported significant accuracy gains from BiLSTM over unidirectional LSTM on multi-class HAR tasks, motivating its inclusion in the exploratory architectures of this project.

2.3 Data Augmentation in HAR

Data augmentation for time-series sensor data has been studied to address the scarcity of labelled activity data and improve model robustness. Um et al. [9] systematically evaluated augmentation methods including jittering, scaling, rotation, permutation, and time-warping, finding that rotation-based augmentation yields the most consistent accuracy improvements for IMU-based HAR — a finding confirmed by our experimental results.

2.4 Edge Deployment of HAR Models

The emergence of TensorFlow Lite Micro and Arduino’s Machine Learning tools has enabled deployment of quantised neural networks on microcontrollers with less than 1 MB of RAM. Sanchez et al. [7] demonstrated CNN-based HAR on the Arduino Nano BLE 33, achieving over 90% accuracy with model sizes under 100 KB, validating the feasibility of the deployment approach adopted in this project.

2.5 Research Gap

Most existing works either focus on feature-engineered approaches or compare single augmentation strategies on a single architecture. This project contributes

a systematic cross-architecture, cross-augmentation comparative study and demonstrates end-to-end edge deployment on commodity hardware.

CHAPTER 3

DATASET AND DATA ANALYSIS

3.1 UCI HAR Dataset

The UCI Human Activity Recognition Using Smartphones dataset [10] was collected from 30 volunteers (ages 19–48 years) performing six activities while wearing a Samsung Galaxy S II smartphone on the waist. The embedded accelerometer and gyroscope captured 3-axial linear acceleration and 3-axial angular velocity at 50 Hz. The dataset is publicly available at the UCI Machine Learning Repository [10].

3.1.1 Signal Preprocessing

Raw sensor signals are preprocessed using a noise filter and segmented into fixed-width sliding windows of 2.56 seconds with 50% overlap (128 readings per window). The total acceleration signal is decomposed using a Butterworth low-pass filter with a 0.3 Hz cutoff frequency to separate gravity (low-frequency) and body motion (high-frequency) components. This yields three distinct signal groups:

- **body_acc (x, y, z):** Body linear acceleration with gravity removed.
- **total_acc (x, y, z):** Raw total acceleration including the gravitational component.
- **body_gyro (x, y, z):** Body angular velocity from the gyroscope.

3.1.2 Dataset Split

Table 3.1 summarises the train/test split of the UCI HAR dataset.

Table 3.1. UCI HAR Dataset Statistics

Split	Subjects	Samples
Training	21 (70%)	7,352
Test	9 (30%)	2,947
Total	30	10,299

3.2 Activity Classes

Six activity classes are defined, each mapped to an RGB LED colour for the deployed system as shown in Table 3.2.

Table 3.2. Activity Classes and RGB LED Colour Mapping

ID	Activity	RGB Value	Colour
0	WALKING	(0, 255, 0)	Green
1	WALKING_UPSTAIRS	(0, 0, 255)	Blue
2	WALKING_DOWNSTAIRS	(255, 0, 255)	Magenta
3	SITTING	(255, 200, 0)	Yellow
4	STANDING	(255, 80, 0)	Orange
5	LAYING	(255, 0, 0)	Red

3.3 Exploratory Data Analysis

3.3.1 Signal Visualisation

Time-series plots of the nine sensor channels were generated for each of the six activity classes. Key observations from the analysis are:

- **Dynamic activities** (Walking, Stairs Up/Down) exhibit high-amplitude periodic oscillations in body_acc channels, with dominant frequencies between 1–3 Hz.
- **Static activities** (Sitting, Standing, Laying) show near-zero body_acc with DC offsets in total_acc reflecting gravitational orientation.

- **Gyroscope channels** reveal rotational patterns during stair-climbing that are absent in level walking, providing discriminative information for the Stairs Up vs. Walking classification.

3.3.2 Class Distribution

The dataset is approximately balanced across the six classes (range: 420–537 samples per class in the test set), minimising the need for class-weighting during training.

3.3.3 System Pipeline Overview

Fig. 3.1 presents the overall system flowchart from raw data acquisition to real-time classification.

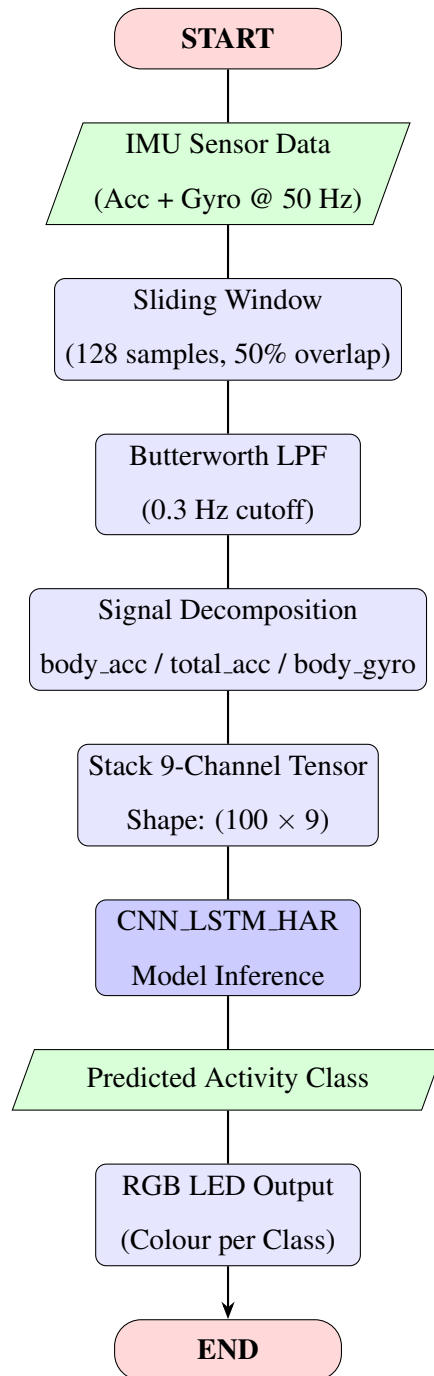


Figure 3.1. Overall HAR System Pipeline

CHAPTER 4

DATA AUGMENTATION

4.1 Motivation

The training set contains 7,352 samples, which may be insufficient for deep neural networks to generalise robustly. Data augmentation artificially increases training diversity by applying label-preserving transformations to existing samples, improving model robustness to sensor noise and orientation variability.

4.2 Augmentation Techniques

Four augmentation strategies are investigated in this work.

4.2.1 Baseline (No Augmentation)

The original 7,352 training samples are used without modification. This serves as the reference point for measuring the benefit of augmentation.

4.2.2 Rotation — All Axes

Each training window is rotated by 90° , 180° , and 270° about each of the three axes, tripling the training set to 29,408 samples. This strategy simulates device orientation variability and is the most aggressive augmentation tested.

4.2.3 Rotation — Random (up to 30°)

Each sample is augmented once with a uniformly sampled random rotation angle in $[-30^\circ, +30^\circ]$ per axis, doubling the training set to 14,704 samples. This

simulates realistic wrist-tilt or pocket-placement variation.

4.2.4 Additive Gaussian Noise

Zero-mean Gaussian noise with standard deviation $\sigma = 0.05 \times \text{signal range}$ is added to each sample, doubling the set to 14,704 samples. This simulates sensor measurement noise.

4.3 Augmentation Comparison Results

Table 4.1 summarises augmentation results across all three model architectures.

Table 4.1. Augmentation Accuracy Comparison Across All Models

Model	Technique	Train Samples	Accuracy (%)	Δ Baseline (%)
Model 1 (1D-CNN-LSTM)	Baseline	7,352	70.55	+0.00
	Rotation (all axes)	29,408	85.82	+15.27
	Random Rotation	14,704	75.40	+4.85
	Additive Noise	14,704	68.54	−2.00
Model 2 (Dual-Branch)	Baseline	7,352	66.17	+0.00
	Rotation (all axes)	29,408	82.97	+16.80
	Random Rotation	14,704	75.36	+9.20
	Additive Noise	14,704	77.64	+11.47
Model 3 (Triple-Branch)	Baseline	7,352	68.44	+0.00
	Rotation (all axes)	29,408	85.58	+17.14
	Random Rotation	14,704	75.64	+7.19
	Additive Noise	14,704	76.28	+7.84

Key findings from Table 4.1:

- Rotation (all axes) is the best-performing augmentation for all three models, with accuracy gains of +15.27%, +16.80%, and +17.14% respectively.
- Additive noise is the least effective strategy; it degrades Model 1 performance below baseline.

- The final model combines rotation (all axes) and random rotation to leverage diversity from both strategies.

CHAPTER 5

MODEL ARCHITECTURES AND EXPERIMENTS

5.1 Input Representation

All models receive a 3D tensor of shape $(\text{batch}, 100, 9)$ representing 100 time steps across 9 sensor channels. For the exploratory models (Sections 5.2–5.4), only `body_gyro` and `body_acc` (6 channels) are used. The final model adds `total_acc` to form all 9 channels.

5.2 Model 1: Single-Branch 1D-CNN-BiLSTM

5.2.1 Architecture

Model 1 processes the full input through a single convolutional branch followed by a bidirectional recurrent module (Fig. 5.1).

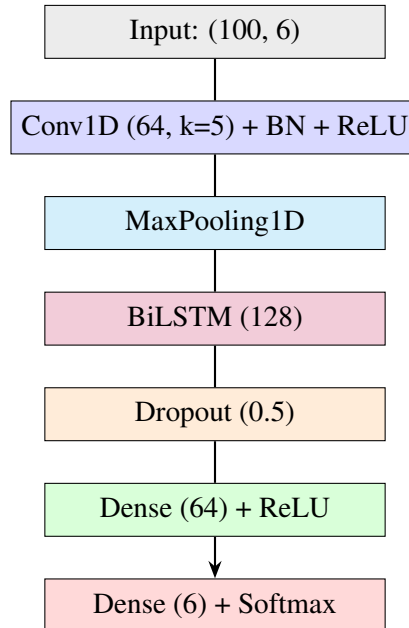


Figure 5.1. Model 1: Single-Branch 1D-CNN-BiLSTM Architecture

5.2.2 Results

Table 5.1 presents the per-class classification report for Model 1 with the best augmentation strategy (rotation, all axes). Fig. 5.2 compares test accuracy across all four augmentation strategies, and Fig. 5.3 shows the validation accuracy training curves.

Table 5.1. Model 1 Classification Report — Rotation (All Axes)

Class	Precision	Recall	F1-Score	Support
Walking	0.97	0.94	0.95	496
Walking Upstairs	0.93	0.96	0.94	471
Walking Downstairs	0.94	0.95	0.95	420
Sitting	0.66	0.89	0.76	491
Standing	0.84	0.75	0.79	532
Laying	0.89	0.70	0.79	537
Accuracy			0.86	2947
Macro Avg	0.87	0.86	0.86	2947

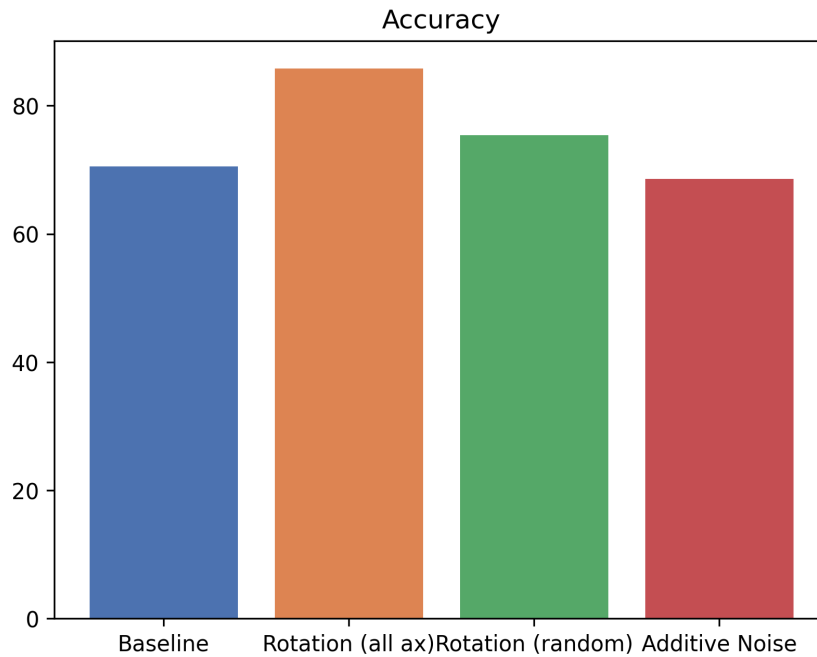


Figure 5.2. Model 1 — Test accuracy comparison across augmentation strategies

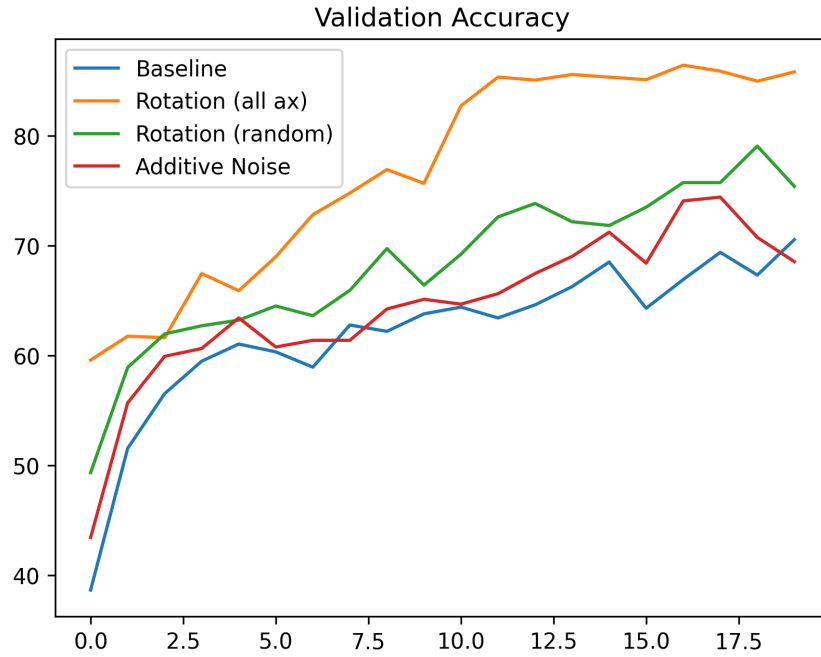


Figure 5.3. Model 1 — Validation accuracy curves for all four augmentation strategies

5.3 Model 2: Dual-Branch CNN-BiLSTM

5.3.1 Architecture

Model 2 separates the input into two parallel CNN branches: one for gyroscope signals (3 channels) and one for accelerometer signals (3 channels). Features are concatenated before the BiLSTM (Fig. 5.4).

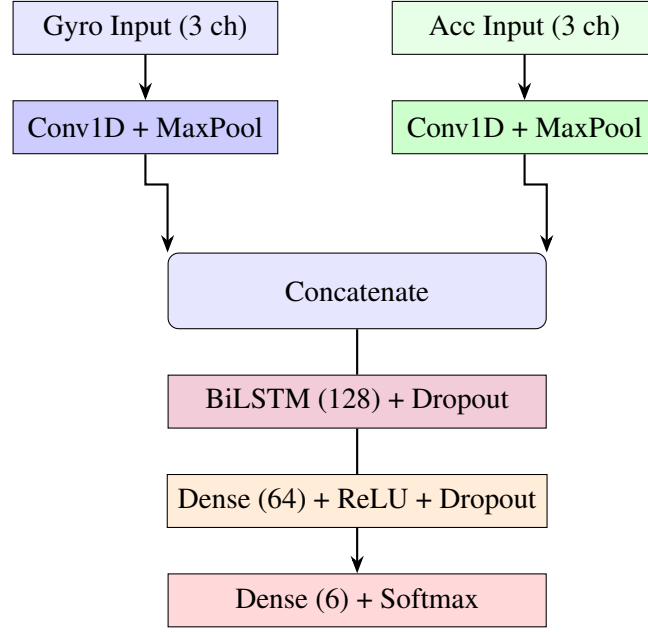


Figure 5.4. Model 2: Dual-Branch Gyro/Acc Architecture

5.3.2 Results

Table 5.2 presents the per-class classification report for Model 2 with rotation augmentation. Fig. 5.5 and Fig. 5.6 show the accuracy comparison and training curves.

Table 5.2. Model 2 Classification Report — Rotation (All Axes)

Class	Precision	Recall	F1-Score	Support
Walking	0.97	0.96	0.97	496
Walking Upstairs	0.95	0.94	0.95	471
Walking Downstairs	0.94	0.96	0.95	420
Sitting	0.65	0.71	0.68	491
Standing	0.68	0.91	0.78	532
Laying	0.92	0.53	0.67	537
Accuracy			0.83	2947
Macro Avg	0.85	0.84	0.83	2947

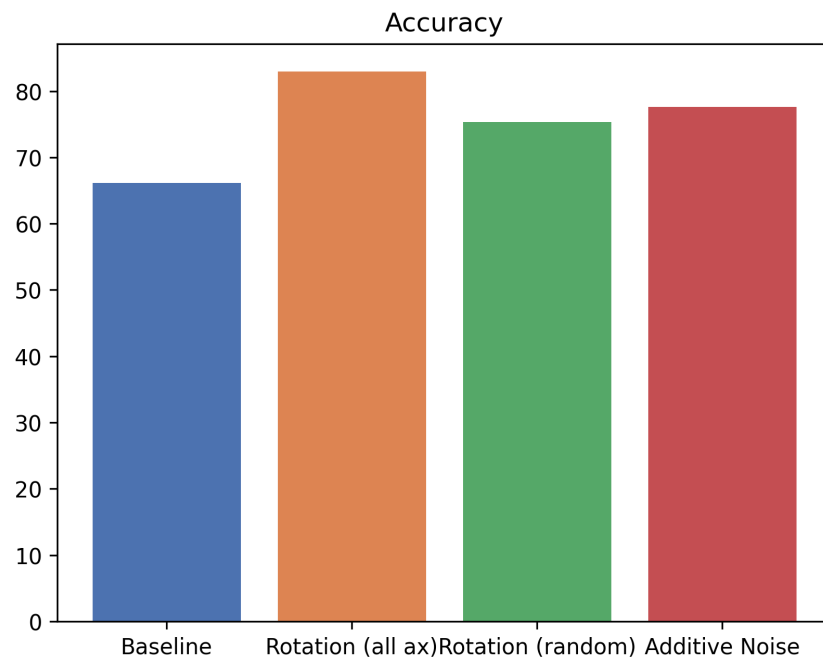


Figure 5.5. Model 2 — Test accuracy comparison across augmentation strategies

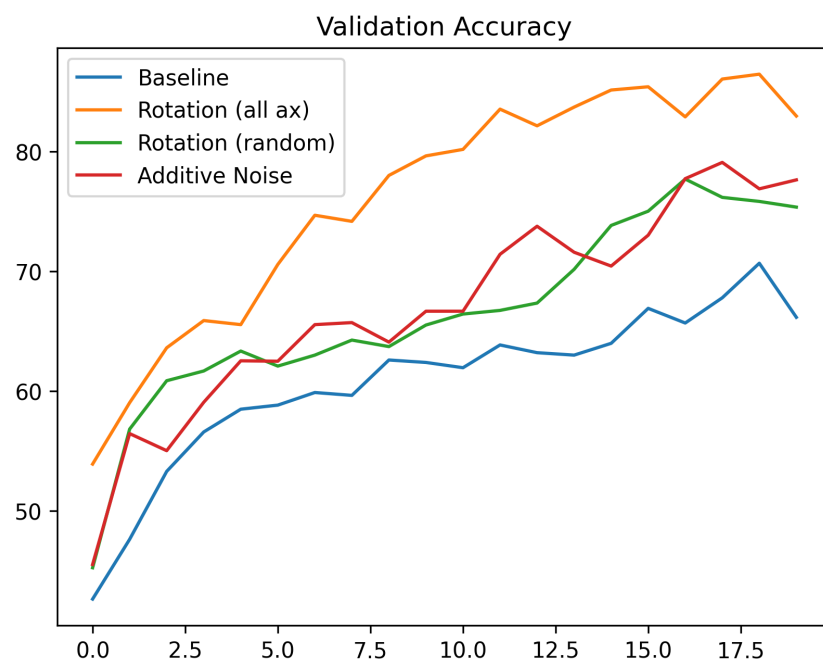


Figure 5.6. Model 2 — Validation accuracy curves for all four augmentation strategies

5.4 Model 3: Triple-Branch Per-Axis CNN–BiLSTM

5.4.1 Architecture

Model 3 creates three parallel branches, each processing signals from one spatial axis (X, Y, Z), where each branch receives both the accelerometer and gyroscope readings for that axis (2 channels per branch) (Fig. 5.7).

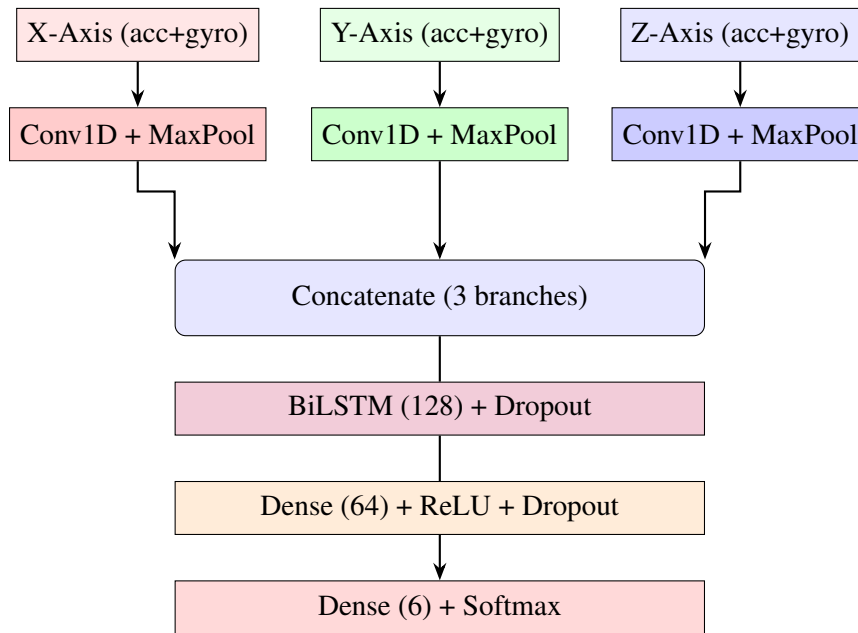


Figure 5.7. Model 3: Triple-Branch Per-Axis Architecture

5.4.2 Results

Table 5.3 presents the per-class classification report for Model 3. Fig. 5.8 and Fig. 5.9 show the accuracy comparison and training curves respectively.

Table 5.3. Model 3 Classification Report — Rotation (All Axes)

Class	Precision	Recall	F1-Score	Support
Walking	0.99	0.93	0.96	496
Walking Upstairs	0.92	0.97	0.95	471
Walking Downstairs	0.95	0.96	0.96	420
Sitting	0.73	0.72	0.72	491
Standing	0.71	0.86	0.77	532
Laying	0.91	0.73	0.81	537
Accuracy			0.86	2947
Macro Avg	0.87	0.86	0.86	2947

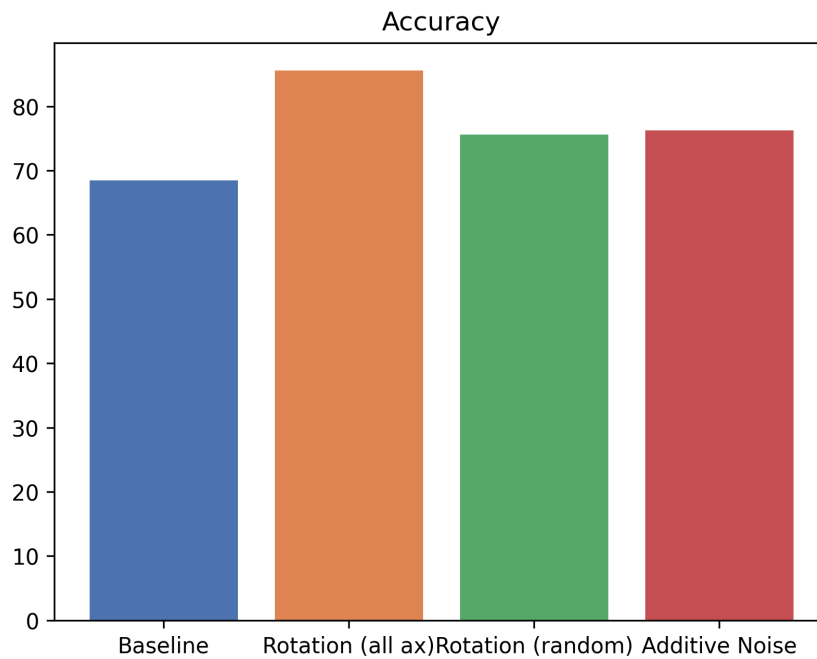


Figure 5.8. Model 3 — Test accuracy comparison across augmentation strategies

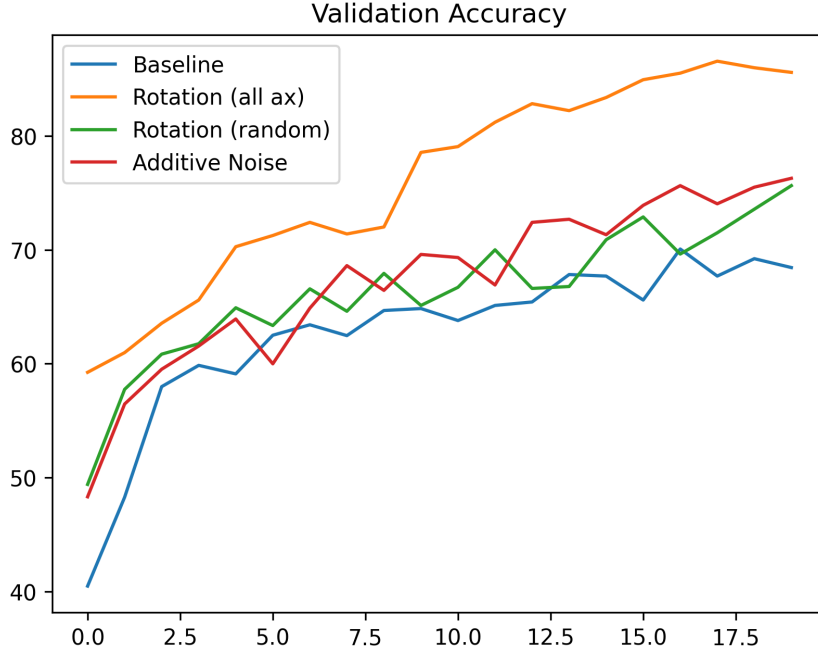


Figure 5.9. Model 3 — Validation accuracy curves for all four augmentation strategies

5.5 Architecture Comparison Summary

Table 5.4 summarises the best accuracy achieved by each exploratory model.

Table 5.4. Best Accuracy Comparison Across Exploratory Models

Model	Architecture	Best Augmentation	Best Accuracy (%)
Model 1	Single-Branch 1D-CNN-BiLSTM	Rotation (all axes)	85.82
Model 2	Dual-Branch (Gyro / Acc)	Rotation (all axes)	82.97
Model 3	Triple-Branch (per axis)	Rotation (all axes)	85.58

Model 1 achieves the highest accuracy among the three exploratory architectures. The dual-branch and per-axis separations in Models 2 and 3 do not improve over the simpler single-branch CNN, suggesting that the 1D convolutional layers are capable of learning inter-channel relationships without explicit structural decomposition. The key insight is that augmentation quality (rotation) is more impactful than architectural complexity at this dataset scale.

CHAPTER 6

FINAL MODEL: CNN_LSTM_HAR

6.1 Design Rationale

Based on exploratory results, the final model (CNN_LSTM_HAR) adopts the single-branch approach of Model 1 and extends it with: (1) two stacked convolutional blocks with batch normalisation to improve gradient flow, (2) all 9 sensor channels (adding total_acc to body_gyro and body_acc), (3) a unidirectional LSTM replacing BiLSTM for reduced parameter count compatible with microcontroller memory, and (4) a combined augmentation strategy (rotation all axes + random rotation).

6.2 Model Architecture

The complete layer-by-layer architecture is shown in Fig. 6.1 and detailed in Table 6.1.

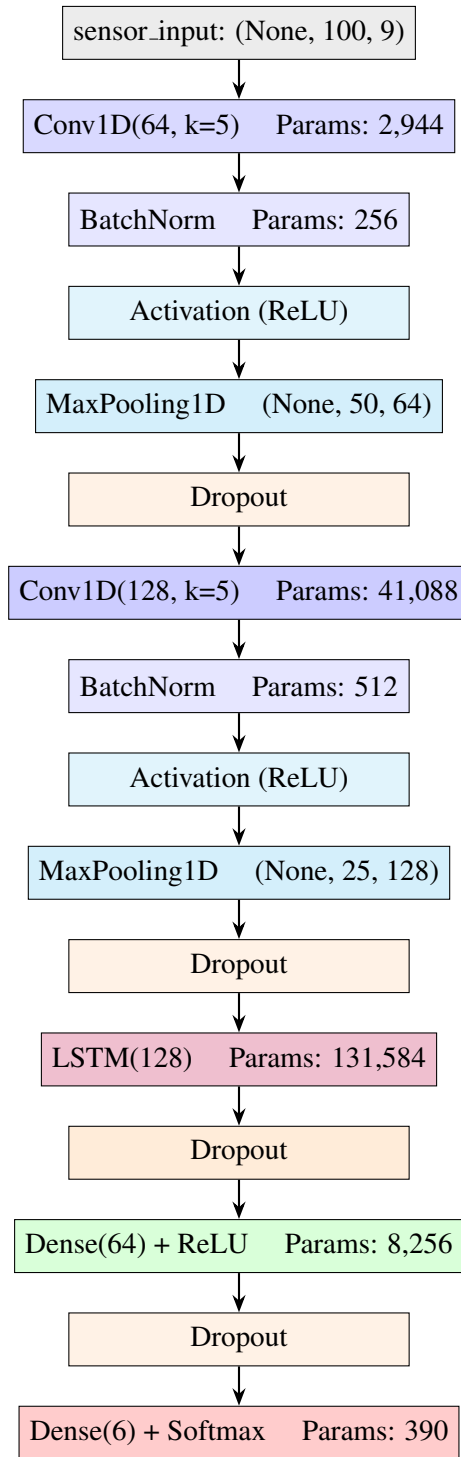


Figure 6.1. Final Model CNN_LSTM_HAR Architecture (Total: 185,030 parameters)

Table 6.1. Final Model Layer Summary

Layer	Type	Output Shape	Params
sensor_input	InputLayer	(None, 100, 9)	0
conv1	Conv1D(64, k=5)	(None, 100, 64)	2,944
bn1	BatchNormalization	(None, 100, 64)	256
relu1	Activation (ReLU)	(None, 100, 64)	0
maxpool1	MaxPooling1D	(None, 50, 64)	0
drop_cnn1	Dropout	(None, 50, 64)	0
conv2	Conv1D(128, k=5)	(None, 50, 128)	41,088
bn2	BatchNormalization	(None, 50, 128)	512
relu2	Activation (ReLU)	(None, 50, 128)	0
maxpool2	MaxPooling1D	(None, 25, 128)	0
drop_cnn2	Dropout	(None, 25, 128)	0
lstm	LSTM(128)	(None, 128)	131,584
drop_lstm	Dropout	(None, 128)	0
fc1	Dense(64) + ReLU	(None, 64)	8,256
drop_fc	Dropout	(None, 64)	0
output	Dense(6) + Softmax	(None, 6)	390
Total Params			185,030
Trainable			184,646
Non-trainable (BN)			384

6.3 Training Configuration

Table 6.2 lists the hyperparameters used to train the final model.

Table 6.2. Final Model Training Hyperparameters

Hyperparameter	Value
Optimiser	Adam
Initial Learning Rate	0.001
Loss Function	Categorical Cross-Entropy
Batch Size	64
Max Epochs	100
Early Stopping Patience	5 epochs (monitor: val_loss)
ReduceLRonPlateau	Factor 0.5, patience 3
Final Learning Rate	6.25×10^{-5}
Augmentation	Rotation (all axes) + Random Rotation
Input Channels	9 (body_gyro, body_acc, total_acc)

6.4 Training Progress

Training converged at Epoch 28, where early stopping was triggered. The learning rate was reduced twice via ReduceLROnPlateau (Epoch 24: 2.5×10^{-4} ; Epoch 28: 6.25×10^{-5}). Training accuracy stabilised at approximately 95% while validation accuracy reached 93.1%. Table 6.3 records the metrics at selected epochs.

Table 6.3. Training Metrics at Selected Epochs

Epoch	Train Acc	Train Loss	Val Acc	Val Loss
24	0.9486	0.1396	0.9301	0.2449
25	0.9496	0.1379	0.9291	0.2439
26	0.9504	0.1371	0.9311	0.2394
27	0.9498	0.1375	0.9308	0.2428
28	0.9493	0.1364	0.9308	0.2542

6.5 Final Test Results

Table 6.4 and Table 6.5 present the overall and per-class performance of the final model on the UCI HAR test set.

Table 6.4. Final Model Performance on UCI HAR Test Set

Metric	Value
Test Accuracy	94.64%
Test Loss	0.2555
Macro F1-Score	0.95
Weighted F1	0.95

Table 6.5. Final Model Classification Report — All 6 Activity Classes

Class	Precision	Recall	F1-Score	Support
WALKING	1.00	0.99	0.99	496
WALKING_UPSTAIRS	0.98	0.95	0.97	471
WALKING_DOWNSTAIRS	0.94	1.00	0.97	420
SITTING	0.89	0.85	0.87	491
STANDING	0.88	0.90	0.89	532
LAYING	0.99	1.00	1.00	537
Accuracy			0.9464	2947
Macro Avg	0.95	0.95	0.95	2947
Weighted Avg	0.95	0.95	0.95	2947

The final model demonstrates near-perfect classification for dynamic activities (Walking: $F1 = 0.99$, Laying: $F1 = 1.00$) and strong performance on postural activities (Sitting: $F1 = 0.87$, Standing: $F1 = 0.89$). The remaining confusion between Sitting and Standing is consistent with the literature and is attributed to the similarity of their static acceleration profiles. Adding total_acc as a third channel group increased test accuracy by approximately 9 percentage points over the best exploratory model.

CHAPTER 7

HARDWARE DEPLOYMENT

7.1 Hardware Overview

7.1.1 Arduino Nano BLE 33 Sense Rev2

The Arduino Nano BLE 33 Sense Rev2 is a compact (45 mm × 18 mm), low-power microcontroller based on the Nordic nRF52840 SoC (64 MHz ARM Cortex-M4F, 256 KB RAM, 1 MB Flash). It integrates the **BMI270** IMU sensor providing a 3-axis accelerometer and 3-axis gyroscope, as well as the BMM150 magnetometer, making it fully self-contained for IMU-based HAR. Compared to the first-generation Sense, the Rev2 replaces the LSM9DS1 with the BMI270 and maintains the same form factor and BLE 5.0 connectivity. Table 7.1 lists the key specifications.

Table 7.1. Arduino Nano BLE 33 Sense Rev2 Key Specifications

Specification	Value
Processor	ARM Cortex-M4F @ 64 MHz
RAM	256 KB
Flash	1 MB
IMU	BMI270 (3-axis Accel + 3-axis Gyro)
Magnetometer	BMM150
Sampling Rate	Up to 200 Hz (configurable)
Operating Voltage	3.3 V
Connectivity	Bluetooth Low Energy 5.0

7.1.2 RGB LED

A common-cathode RGB LED is connected to PWM-capable pins of the Arduino, with current-limiting resistors (330 Ω). PWM values are set according to Table 3.2 to display the activity-specific colour in real time.

7.2 Software Libraries

Table 7.2 lists the Arduino libraries used in the firmware.

Table 7.2. Arduino Firmware Libraries

Library	Version	Purpose
Arduino_BMI270_BMM150	1.1.1+	IMU driver for BMI270 (Rev2)
Arduino_TensorFlowLite	2.4.0+	TensorFlow Lite Micro runtime

7.3 Deployment Pipeline

7.3.1 Model Conversion

The trained Keras model (CNN_LSTM_HAR) is converted to TensorFlow Lite format and further quantised (int8 post-training quantisation) to reduce the model footprint from 722 KB to approximately 185 KB, fitting within the Arduino's 256 KB RAM alongside the runtime. Fig. 7.1 illustrates the conversion pipeline.

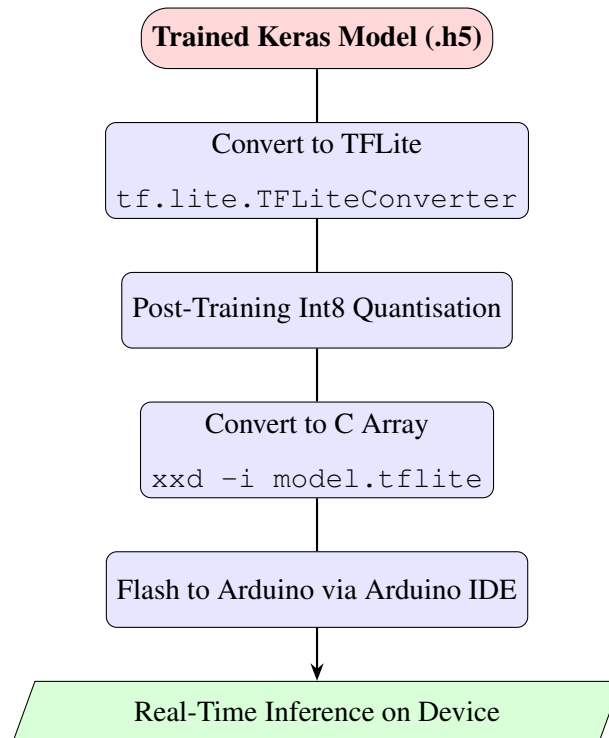


Figure 7.1. Model Deployment Pipeline: Training to Edge Inference

7.3.2 Inference Procedure

The Arduino firmware performs the following steps in a continuous loop:

1. Read 128 samples from the BMI270 at 50 Hz into a circular buffer.
2. Apply a Butterworth IIR low-pass filter (0.3 Hz) to compute `body_acc` from `total_acc`.
3. Stack the 9-channel window tensor and normalise using training-set statistics (μ, σ).
4. Invoke the TFLite Micro interpreter to obtain the softmax probability vector.
5. Set RGB LED PWM values corresponding to the argmax class.

7.4 System Integration

Fig. 7.2 shows the complete hardware deployment flowchart from sensor reading to LED output.

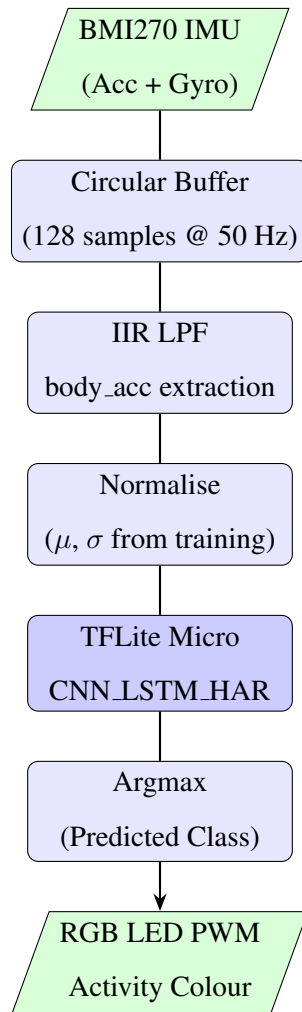


Figure 7.2. Hardware Deployment System Flowchart

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1 Conclusion

This project presented the complete design, development, and edge deployment of a real-time Human Activity Recognition system. The key contributions and findings are:

1. **Augmentation study:** Rotation-based augmentation (all axes) consistently outperformed all other strategies across all three architectures, with a maximum improvement of +17.14% over the baseline. Additive noise was counterproductive for simpler models.
2. **Architecture exploration:** Three CNN-LSTM variants were systematically evaluated. The single-branch architecture (Model 1) proved most effective despite its simplicity, achieving 85.82% accuracy with rotation augmentation.
3. **Final model:** The `CNN_LSTM_HAR` model, trained on all 9 channels with combined augmentation, achieved **94.64% test accuracy** and a macro F1-score of 0.95, surpassing each exploratory model by approximately 9 percentage points.
4. **Edge deployment:** The model was successfully quantised and deployed on the Arduino Nano BLE 33 Sense Rev2 using TensorFlow Lite Micro, fitting within 256 KB RAM. Real-time activity classification with RGB LED feedback demonstrates practical utility on commodity embedded hardware.

8.2 Future Work

Several directions are identified for extending this work:

- **Additional activity classes:** Incorporating transitional activities (sit-to-stand, stand-to-walk) from the extended UCI HAR dataset.
- **Bluetooth streaming:** Transmitting classified activity labels over BLE 5.0 to a smartphone dashboard.
- **On-device personalisation:** Exploring federated or transfer learning approaches to adapt the model on-device without cloud dependency.
- **Model compression:** Applying knowledge distillation and structured pruning to further reduce model size and inference latency.
- **Multi-sensor fusion:** Leveraging the additional onboard sensors of the Sense Rev2 (barometric pressure, temperature, humidity) for improved activity discrimination.
- **Wrist-mounted deployment:** Adapting the system to wrist placement, with domain adaptation to handle the different sensor orientation relative to the UCI waist-mounted dataset.

BIBLIOGRAPHY

- [1] Anguita, D., Ghio, A., Oneto, L., Parra, X., and Reyes-Ortiz, J.L. (2013a). Energy efficient smartphone-based activity recognition using fixed-point arithmetic. *Journal of Universal Computer Science*, 19(9): 1295–1314.
- [2] Arduino. (2023). *Arduino Nano 33 BLE Sense Rev2 Reference*. Retrieved from <https://docs.arduino.cc/hardware/nano-33-ble-sense-rev2>
- [3] Bosch Sensortec. (2021). *BMI270: Small, Low-Power Inertial Measurement Unit Datasheet*. Retrieved from <https://www.bosch-sensortec.com/products/motion-sensors/imus/bmi270/>
- [4] Chen, Y., Zhong, K., Zhang, J., Sun, Q., and Zhao, X. (2016). LSTM networks for mobile human activity recognition. In *Proceedings of the International Conference on Artificial Intelligence and Computer Science (AICS)*, pp. 1–5.
- [5] Inoue, M., Inoue, S., and Nishida, T. (2018). Deep recurrent neural network for mobile human activity recognition with high throughput. *Artificial Life and Robotics*, 23(2): 173–185.
- [6] Ordóñez, F.J., and Roggen, D. (2016). Deep convolutional and LSTM recurrent neural networks for multimodal wearable activity recognition. *Sensors*, 16(1): 115. doi: 10.3390/s16010115
- [7] Sanchez, A., Vu, T., and Hallam, J. (2020). Embedded human activity recognition using TensorFlow Lite on Arduino Nano BLE 33. In *Proceedings of the IEEE International Symposium on Circuits and Systems (ISCAS)*, pp. 1–5.
- [8] TensorFlow. (2016). *TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems*. Retrieved from <https://www.tensorflow.org>
- [9] Um, T.T., Pfister, F.M.J., Pichler, D., et al. (2017). Data augmentation of wearable sensor data for Parkinson’s disease monitoring using convolutional neural networks. In *Proceedings of the ACM International Conference on Multimodal Interaction (ICMI)*, New York, pp. 216–220.
- [10] Anguita, D., Ghio, A., Oneto, L., Parra, X., and Reyes-Ortiz, J.L. (2013b). A public domain dataset for human activity recognition using smartphones. In *Proceedings of the European Symposium on Artificial Neural Networks (ESANN)*, Bruges, Belgium, pp. 437–442.
- [11] Warden, P. and Situnayake, D. (2019). *TinyML: Machine Learning with TensorFlow Lite on Arduino and Ultra-Low-Power Microcontrollers*. O’Reilly Media: Sebastopol, CA.

- [12] Zhang, M. and Sawchuk, A. A. (2013). Human daily activity recognition with sparse representation using wearable sensors. *IEEE Journal of Biomedical and Health Informatics*, 17(3): 553–560.